

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{dx}{dy} = 8 \cos 2y \Rightarrow \frac{dy}{dx} = \frac{1}{8 \cos 2y}$	M1	This mark is given for differentiating and inverting
	At (0, 0), $\frac{dy}{dx} = \frac{1}{8}$	A1	This mark is given for finding $\frac{dy}{dx}$ when $y = 0$
(b)(i)	$\sin 2y \approx 2y \Rightarrow x \approx 8y$	B1	This mark is given for finding an approximation for x
(b)(ii)	When x and y are small, $x = 4 \sin 2y$ approximates to the line $x = 8y$	B1	This mark is given for a valid explanation of the relationship between x and y when both are small
(c)	$\sin^2 2y + \cos^2 2y = 1$ $\Rightarrow \cos^2 2y = 1 - \sin^2 2y$ $x = 4 \sin 2y \Rightarrow \sin^2 2y = \left(\frac{x}{4}\right)^2$	M1	This mark is given for a method to use find an expression for $\sin^2 2y$ in terms of x
	$\frac{dy}{dx} = \frac{1}{8 \cos 2y} = \frac{1}{8 \sqrt{1 - \left(\frac{x}{4}\right)^2}}$	A1	This mark is given for an unsimplified expression for $\frac{dy}{dx}$
	$\frac{dy}{dx} = \frac{1}{2\sqrt{16 - x^2}}$	A1	This mark is given for a fully correct answer with $a = 2$ and $b = 16$

Q2.

Question	Scheme	Marks	AOs
	Attempts either $\sin 3\theta \approx 3\theta$ or $\cos 4\theta \approx 1 - \frac{(4\theta)^2}{2}$ in $\frac{1 - \cos 4\theta}{2\theta \sin 3\theta}$	M1	1.1b
	Attempts both $\sin 3\theta \approx 3\theta$ and $\cos 4\theta \approx 1 - \frac{(4\theta)^2}{2} \rightarrow \frac{1 - \left(1 - \frac{(4\theta)^2}{2}\right)}{2\theta \times 3\theta}$ and attempts to simplify	M1	2.1
	$= \frac{4}{3}$ oe	A1	1.1b
		(3)	
(3 marks)			

M1: Attempts either $\sin 3\theta \approx 3\theta$ or $\cos 4\theta \approx 1 - \frac{(4\theta)^2}{2}$ in the given expression.

See below for description of marking of $\cos 4\theta$

M1: Attempts to substitute both $\sin 3\theta \approx 3\theta$ and $\cos 4\theta \approx 1 - \frac{(4\theta)^2}{2}$

$$\rightarrow \frac{1 - \left(1 - \frac{(4\theta)^2}{2}\right)}{2\theta \times 3\theta} \text{ and attempts to simplify.}$$

Condone missing bracket on the 4θ so $\cos 4\theta \approx 1 - \frac{4\theta^2}{2}$ would score the method

Expect to see it simplified to a single term which could be in terms of θ

Look for an answer of k but condone $k\theta$ following a slip

A1: Uses both identities and simplifies to $\frac{4}{3}$ or exact equivalent with no incorrect lines BUT allow

recovery on missing bracket for $\cos 4\theta \approx 1 - \frac{4\theta^2}{2}$.

Eg. $\frac{1 - \left(1 - \frac{(4\theta)^2}{2}\right)}{2\theta \times 3\theta} = \frac{8\theta^3}{6\theta} = \frac{4}{3}$ is M1 M1 A0

Condone awrt 1.33.

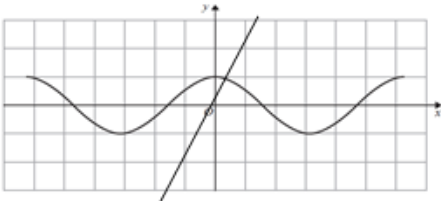
Alt: $\frac{1 - \cos 4\theta}{2\theta \sin 3\theta} = \frac{1 - (1 - 2\sin^2 2\theta)}{2\theta \sin 3\theta} = \frac{2\sin^2 2\theta}{2\theta \sin 3\theta} = \frac{2 \times (2\theta)^2}{2\theta \times 3\theta} = \frac{4}{3}$

M1 For an attempt at $\sin 3\theta \approx 3\theta$ or the identity $\cos 4\theta = 1 - 2\sin^2 2\theta$ with $\sin 2\theta \approx 2\theta$

M1 For both of the above and attempts to simplify to a single term.

A1 $\frac{4}{3}$ oe

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)		M1	This mark is given for plotting the line $y = 2x + \frac{1}{2}$ on the diagram with a correct gradient and intercept
	Only one intersection means that there is one root	A1	This mark is given for a reason why there is only one real root
(b)	$1 - \frac{x^2}{2} - 2x - \frac{1}{2} = 0$	M1	This mark is given for using the small angle approximation $\cos x = 1 - \frac{x^2}{2}$ in the given equation
	$x^2 + 4x - 1 = 0$	M1	This mark is given for rearranging to find a quadratic equation to solve
	0.236 or $-2 + \sqrt{5}$	A1	This mark is given for finding the correct (positive) solution for x

Q4.

Question	Scheme	Marks	AOs
	Examples: $4 \sin \frac{\theta}{2} \approx 4 \left(\frac{\theta}{2} \right), \quad 3 \cos^2 \theta \approx 3 \left(1 - \frac{\theta^2}{2} \right)^2$ $3 \cos^2 \theta = 3(1 - \sin^2 \theta) \approx 3(1 - \theta^2)$ $3 \cos^2 \theta = 3 \frac{(\cos 2\theta + 1)}{2} \approx \frac{3}{2} \left(1 - \frac{4\theta^2}{2} + 1 \right)$	M1	1.1a
	Examples: $4 \sin \frac{\theta}{2} + 3 \cos^2 \theta \approx 4 \left(\frac{\theta}{2} \right) + 3 \left(1 - \frac{\theta^2}{2} \right)^2$ $4 \sin \frac{\theta}{2} + 3 \cos^2 \theta = 4 \left(\frac{\theta}{2} \right) + 3(1 - \sin^2 \theta) \approx 2\theta + 3(1 - \theta^2)$ $4 \sin \frac{\theta}{2} + 3 \cos^2 \theta = 4 \sin \frac{\theta}{2} + 3 \frac{(\cos 2\theta + 1)}{2} \approx 4 \left(\frac{\theta}{2} \right) + \frac{3}{2} \left(1 - \frac{4\theta^2}{2} + 1 \right)$	dM1	1.1b
	$= 2\theta + 3(1 - \theta^2 + \dots) = 3 + 2\theta - 3\theta^2$	A1	2.1
		(3)	
(3 marks)			
Notes			
M1: Attempts to use at least one correct approximation within the given expression. Either $\sin \frac{\theta}{2} \approx \frac{\theta}{2}$ or $\cos \theta \approx 1 - \frac{\theta^2}{2}$ or e.g. $\sin \theta \approx \theta$ if they write $\cos^2 \theta$ as $1 - \sin^2 \theta$ or e.g. $\cos 2\theta \approx 1 - \frac{(2\theta)^2}{2}$ (condone missing brackets) if they write $\cos^2 \theta$ as $\frac{1 + \cos 2\theta}{2}$. Allow sign slips only with any identities used but the appropriate approximations must be applied. dM1: Attempts to use correct approximations with the given expression to obtain an expression in terms of θ only. Depends on the first method mark. A1: Correct terms following correct work. Allow the terms in any order and ignore any extra terms if given correct or incorrect.			

Q5.

Question	Scheme	Marks	AOs
(a)	States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \times \frac{\cos \theta}{\sin \theta} = \cos \theta \cot \theta$ *	A1*	2.1
		(3)	
(b)	$\operatorname{cosec} x - \sin x = \cos x \cot(3x - 50^\circ)$ $\Rightarrow \cos x \cot x = \cos x \cot(3x - 50^\circ)$		
	$\cot x = \cot(3x - 50^\circ) \Rightarrow x = 3x - 50^\circ$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $\cot x = \cot(3x - 50^\circ) \Rightarrow x + 180^\circ = 3x - 50^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $x = 90^\circ$	B1	2.2a
		(5)	
(8 marks)			
Notes:			

(a) Condone a full proof in x (or other variable) instead of θ 's here

B1: States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$ Do not accept $\operatorname{cosec} \theta = \frac{1}{\sin}$ with the θ missing

M1: For the key step in forming a single fraction/common denominator

E.g. $\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$. Allow if written separately $\frac{1}{\sin \theta} - \sin \theta = \frac{1}{\sin \theta} - \frac{\sin^2 \theta}{\sin \theta}$

Condone missing variables for this M mark

A1*: Shows careful work with all necessary steps shown leading to given answer. See scheme for necessary steps. There should not be any notational or bracketing errors.

(b) Condone θ 's instead of x 's here

M1: Uses part (a), cancels or factorises out the $\cos x$ term, to establish that one solution is found when

$$x = 3x - 50^\circ.$$

You may see solutions where $\cot A - \cot B = 0 \Rightarrow \cot(A - B) = 0$ or $\tan A - \tan B = 0 \Rightarrow \tan(A - B) = 0$.

As long as they don't state $\cot A - \cot B = \cot(A - B)$ or $\tan A - \tan B = \tan(A - B)$ this is acceptable

A1: $x = 25^\circ$

M1: For the key step in realising that $\cot x$ has a period of 180° and a second solution can be found by solving $x + 180^\circ = 3x - 50^\circ$. The sight of $x = 115^\circ$ can imply this mark provided the step $x = 3x - 50^\circ$ has been seen. Using reciprocal functions it is for realising that $\tan x$ has a period of 180°

A1: $x = 115^\circ$ Withhold this mark if there are additional values in the range $(0, 180)$ but ignore values outside.

B1: Deduces that a solution can be found from $\cos x = 0 \Rightarrow x = 90^\circ$. Ignore additional values here.

Solutions with limited working. The question demands that candidates show all stages of working.

SC: $\cos x \cot x = \cos x \cot(3x - 50^\circ) \Rightarrow \cot x = \cot(3x - 50^\circ) \Rightarrow x = 25^\circ, 115^\circ$

They have shown some working so can score B1, B1 marked on open as 11000

Alt 1- Right hand side to left hand side

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$	B1	1.2
	$\cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta} = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{1}{\sin \theta} - \sin \theta = \operatorname{cosec} \theta - \sin \theta$ *	A1*	2.1
		(3)	

Alt 2- Works on both sides

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$ or $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$LHS = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta}$ $RHS = \cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta}$	M1	2.1
	States a conclusion E.g. "HENCE TRUE", "QED" or $\operatorname{cosec} \theta - \sin \theta \equiv \cos \theta \cot \theta$ o.e. (condone = for $\equiv$)	A1*	2.1
		(3)	

Alt (b)

Question	Scheme	Marks	AOs
	$\cot x = \cot(3x - 50^\circ) \Rightarrow \frac{\cos x}{\sin x} = \frac{\cos(3x - 50^\circ)}{\sin(3x - 50^\circ)}$ $\sin(3x - 50^\circ) \cos x - \cos(3x - 50^\circ) \sin x = 0$ $\sin((3x - 50^\circ) - x) = 0$ $2x - 50^\circ = 0$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $2x - 50^\circ = 180^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $\cos x = 0 \Rightarrow x = 90^\circ$	B1	2.2a
		(5)	

Q6.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2
	$(\text{Area} =) \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area} =) \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt *$	A1*	2.1*
	(3)		
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = [-20 \cos^3 t]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 *$	A1*	2.1
	(3)		
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
	(5)		
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $A \int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = ...$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form **A1:** Correct quartic **M1:** Solves quartic **A1:** Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. **Units are required.**

Q7.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{5}$	B1	1.1b
	$\tan \alpha = 2 \Rightarrow \alpha = \dots$	M1	1.1b
	$\alpha = 1.107$	A1	1.1b
		(3)	
	$\theta = 5 + \sqrt{5} \sin\left(\frac{\pi t}{12} + 1.107 - 3\right)$		
(b)	$(5 + \sqrt{5})^\circ\text{C}$ or awrt 7.24°C	B1ft	2.2a
		(1)	
(c)	$\frac{\pi t}{12} + 1.107 - 3 = \frac{\pi}{2} \Rightarrow t =$	M1	3.1b
	$t =$ awrt 13.2	A1	1.1b
	Either 13:14 or 1:14 pm or 13 hours 14 minutes after midnight.	A1	3.2a
		(3)	
(7 marks)			
Notes:			



(a)

B1: $R = \sqrt{5}$ only.**M1:** Proceeds to a value of α from $\tan \alpha = \pm 2$, $\tan \alpha = \pm \frac{1}{2}$, $\sin \alpha = \pm \frac{2}{\sqrt{5}}$ OR $\cos \alpha = \pm \frac{1}{\sqrt{5}}$

It is implied by either awrt 1.11 (radians) or 63.4 (degrees)

A1: $\alpha =$ awrt 1.107

(b)

B1ft: Deduces that the maximum temperature is $(5 + \sqrt{5})^\circ\text{C}$ or awrt 7.24°C Remember to isw
Condone a lack of units. Follow through on their value of R so allow $(5 + R)^\circ\text{C}$

(c)

M1: An complete strategy to find t from $\frac{\pi t}{12} \pm 1.107 - 3 = \frac{\pi}{2}$.

Follow through on their 1.107 but the angle must be in radians.

It is possible via degrees but only using $15t \pm 63.4 - 171.9 = 90$ **A1:** awrt $t = 13.2$ **A1:** The question asks for the time of day so accept either 13:14, 1:14 pm, 13 hours 14 minutes after midnight, 13h 14, or 1 hour 14 minutes after midday. If in doubt use review

It is possible to attempt parts (b) and (c) via differentiation but it is unlikely to yield correct results.

$$\frac{d\theta}{dt} = \frac{\pi}{12} \cos\left(\frac{\pi t}{12} - 3\right) - \frac{2\pi}{12} \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ scores M1 A1 A1}$$

$$\frac{d\theta}{dt} = \cos\left(\frac{\pi t}{12} - 3\right) - 2 \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ they can score M1 A0 A1 (SC)}$$

A value of $t = 1.23$ implies the minimum value has been found and therefore incorrect method M0.